

Raycasting Ground Truth Validation

Synthetic LiDAR for Tool Step 6 (M3C2) Verification

Initial State (Phase A/B/C v1)

- Scripts targeted imaginary tunnel (straight, $R=3.0\text{m}$, 1 scanner, -7/-5/-15mm)
- Never actually run end-to-end
- Config copy-paste risks ("must match byte-for-byte")
- Real dataset blender_lidar_t0t5 is curved ($R=500\text{m}$), 3 TLS stations, progressive T1-T5

Realign to Real Geometry

- Verify .blend via Blender MCP: R=500m curve, radius 4.256m, 3 stations @ arc-len 10/40/70m
- Build single engine (raycast_tunnel_epochs.py) for T0 & Tn — same code path, no drift
- Deform in local cross-section frame (ground_truth.csv magnitudes: crown -45, side -35, local -40mm @ T5)

Single Engine, Two Validators

Engine

- tools/raycast_tunnel_epochs.py
- Blender bpy + BVHTree
- T0 clean, Tn deformed
- Same 3 stations & noise
- Export T0-T5 .txt/.json

Validator 1

- phase_c_validate.py
- Window-mean in frame
- Checks raycast fidelity
- Lower bound (bias low)
- Regenerable

Validator 2

- crosscheck_tool_step6.py
- Call Step 6 M3C2 (prod)
- Independent check
- Mm-level accuracy
- FINAL TRUTH

Validator 1: Window-Mean (Raycast Fidelity)

Epoch	Crown err	Side err	Local err	MAE
T1	0.3	0.0	0.0	0.1mm
T2	0.2	0.0	0.0	0.1mm
T3	0.1	0.1	2.2	0.8mm
T4	1.4	1.6	3.6	2.2mm
T5	1.9	2.3	5.5	3.2mm

Window-mean is lower bound; narrow features read low

Validator 2: Tool M3C2 (py4dgeo) — FINAL TRUTH

Epoch	Crown err	Side err	Local err	Peak
T1	1.0	0.0	0.0	M3C2
T2	0.9	0.4	0.0	M3C2
T3	0.8	0.7	1.0	M3C2
T4	1.4	0.1	1.3	M3C2
T5	1.3	0.1	2.4	M3C2

Tool recovers GT ≤ 2.4 mm peak — mm-level accurate

Ground Truth by Epoch

Crown Settlement @ 20m

T0: 0, T1: -5, T2: -12, T3: -20, T4: -30, T5: -45 mm

Sidewall Convergence @ 45m

T0: 0, T1: 0, T2: -5, T3: -12, T4: -22, T5: -35 mm

Local Damage @ 65m

T0: 0, T1: 0, T2: 0, T3: -15, T4: -25, T5: -40 mm

Technical Validation

- ✓ Geometry verified via Blender MCP inspection (R=500, radius 4.25m, curve centerline)
- ✓ Path robustness: scripts resolve via __file__ (run from any cwd)
- ✓ Full series T0-T5 generated (112k-112.5k points/epoch, ~4.2MB each)
- ✓ Two independent validators (window-mean + tool M3C2)
- ✓ 25MB regenerable point clouds gitignored; metadata/CSV tracked

Conclusions

- Tool Step 6 (py4dgeo M3C2) is mm-level accurate on curved tunnel geometry
- Raycasting ground truth validates the measurement pipeline end-to-end
- Single engine architecture eliminates config-drift risk
- Ready for next phase: higher ray density (0.5°) or real-world validation

Delivery Summary

Phase A

Clean T0
(engine verified)

Phase B

Deformed T1-T5
(series generated)

Phase C

Validation
(2 validators, PASS)

Scripts: 3 main (engine, 2 validators) + plotter + cross-check | Data: T0-T5 JSON, validation CSV, series plot